

Kernel-Preconditioned Burer–Monteiro Flow: A Spectral Implicit Bias for Wide Networks Trained with Triplet Loss

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Abstract

We show that gradient flow on a wide ReLU embedding network, trained with a twice-differentiable loss on its Gram matrix $G = FF^\top$, evolves under a *kernel-preconditioned Burer–Monteiro flow* $\dot{G} = -2(KEG + GEK)$, where $E = \partial L / \partial G$ and K is the depth- L neural tangent kernel at the inputs. When $K = I$ this recovers the classical Burer–Monteiro dynamics for low-rank semidefinite optimization. Beyond this equivalence, the kernel form yields a *spectral implicit bias*: in the eigenbasis of K , mode (i, j) of G evolves at instantaneous rate $\lambda_i + \lambda_j$, and at NTK initialization the per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ scales as λ_i^2 . This is a sharp, falsifiable scaling law that does not follow from the NTK ODE without the BM viewpoint. We verify the slope-2 prediction across 30 multi-seed configurations: synthetic spheres ($N \in \{30, 100, 1000\}$, depth $L \in \{1, 2\}$) yielding fitted slopes 1.38 to 2.19; Fashion-MNIST (1.69 ± 0.03); frozen-feature CIFAR-10 (2.06 ± 0.01 , the closest match to the prediction); and end-to-end CNN training on CIFAR-10 (1.15 ± 0.07 , with the empirical NTK). As an algorithmic consequence, applying K^{-1} to the embedding-side gradient cancels the kernel preconditioner: a one-line patch to backprop flattens the synthetic slope from 2.00 ± 0.11 to 0.74 ± 0.09 and the head-only CIFAR-10 slope from 2.06 to 1.59. On the end-to-end CNN the slope drops from 1.15 ± 0.07 to 0.97 ± 0.11 . On hard-negative CIFAR-10 retrieval, where the loss gradient deliberately loads on K ’s bottom eigenspace, this preconditioner improves Recall@5 by +2.2 pp, Recall@10 by +3.2 pp, and Recall@20 by +2.2 pp over vanilla training (paired t -test $p = 0.004$ at R@10).

1 Introduction

Neural metric learning trains an embedding $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k$ so that $\|f_\theta(x) - f_\theta(y)\|^2$ encodes a desired similarity geometry. Two theoretical frameworks bear on this setting from very different angles. The *neural tangent kernel* [Jacot et al., 2018, Lee et al., 2019] describes sufficiently wide networks as kernel methods governed by a fixed kernel Θ , and yields linear ODEs for the network’s predictions under gradient flow. The *Burer–Monteiro factorization* [Burer and Monteiro, 2003, 2005], central to low-rank semidefinite programming, parameterizes a positive semidefinite matrix X as $X = UU^\top$ and uses U as the optimization variable; the resulting gradient flow on X takes the form $\dot{X} = -2(EX + XE)$.

The two frameworks are usually treated in isolation. We connect them by observing that an embedding network with twice-differentiable loss on $G = FF^\top$ is a Burer–Monteiro factorization

in a non-linear feature space, and that the NTK enters as an explicit *kernel preconditioner* on the BM dynamics. Concretely, in the NTK regime the network’s Gram matrix evolves according to

$$\dot{G} = -2(KEG + GEK), \quad E = \partial L / \partial G, \quad (1)$$

which reduces to the classical BM flow when $K = I$. We call (1) the *kernel-preconditioned Burer–Monteiro* (kBM) flow.

The equivalence is itself a useful conceptual bridge, but on its own it does not say more than the standard NTK ODE in different coordinates. The contribution of this paper is to use the kBM viewpoint to extract a *spectral implicit bias* that does not follow from NTK alone. Diagonalizing $K = U\Lambda U^\top$ and writing $\tilde{G} := U^\top G U$, $\tilde{E} := U^\top E U$, the kBM flow becomes

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

so the time derivative of mode (i, j) is multiplied by $\lambda_i + \lambda_j$. Modes corresponding to small kernel eigenvalues are essentially frozen. This makes a sharp, testable prediction: the trajectory of $G(t)$ confines its motion to K ’s top eigenspaces, regardless of where the loss landscape would otherwise prefer it to go. Under NTK initialisation the prediction sharpens to a *quadratic* scaling: per-mode displacement at any fixed training horizon scales as λ_i^2 , which we verify across synthetic and real-image setups.

Contributions.

1. **Equivalence theorem.** Under gradient flow on a single-layer ReLU network with the standard NTK approximation, the Gram-matrix dynamics take the kBM form (1), recovering the classical BM dynamics when $K = I$ (Theorem 2).
2. **Depth- L extension.** The same kBM equation governs a depth- L fully-connected ReLU network, with K replaced by the depth- L NTK from the recursive Jacot construction (Theorem 4).
3. **Spectral implicit-bias theorems.** In the eigenbasis of K the kBM flow’s (i, j) component evolves at rate $\lambda_i + \lambda_j$; under a benign-loss assumption this implies a model-free linear-in- λ time-constant bound (Theorem 6). Within the NTK regime, where G_0 and K share approximate eigenstructure, the bound tightens to a quadratic-in- λ prediction (Theorem 9), which matches the empirical slope ≈ 2 across configurations.
4. **Funk–Hecke alignment.** For rotationally-invariant inputs on the sphere, the Funk–Hecke theorem identifies K and the initial Gram G_0 as simultaneously diagonal in the spherical-harmonic basis, with a diagonal-entry ratio $c_l \in [c_{\min}, c_{\max}]$ uniformly bounded across frequencies (Proposition 8). This is the structural fact that upgrades the linear bound to the quadratic prediction.
5. **Numerical illustration.** We verify the equivalence and the spectral-bias prediction with multi-seed experiments on synthetic data ($N \in \{10, 30, 100, 1000\}$, depth $L \in \{1, 2, 3\}$), Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN trained from scratch. Empirical power-law slopes range from 1.15 to 2.19 across configurations, all consistent with the bound ≤ 2 from Theorem 9.

6. **Algorithmic payoff: spectral preconditioner.** Applying K^{-1} to the embedding-side gradient flattens the spectral bias from slope 1.99 ± 0.11 (vanilla) to 0.74 ± 0.10 (preconditioned) on synthetic data and from 2.06 to 1.59 on frozen-feature CIFAR-10, a one-line modification to backpropagation derived directly from the kBM viewpoint (Section 4.5). On the end-to-end CNN the slope drops from 1.15 ± 0.07 to 0.97 ± 0.11 .
7. **Test-time gain on hard triplets.** On held-out CIFAR-10 retrieval with hard nearest-different-class triplets, the preconditioner improves recall@10 by $+3.2 \pm 1.1$ percentage points over vanilla (paired t -test $p = 0.004$, Table 2), with R@5 and R@20 gains of $+2.2$ pp each. The gain is concentrated where the gradient signal loads on K ’s bottom eigenspace—exactly as Theorem 9 predicts.
8. **Distinct from Razin–Cohen rank bias.** Our spectral bias operates along K ’s eigenbasis; the Razin–Cohen rank bias operates on G ’s singular spectrum. We verify empirically that vanilla NTK metric learning produces a trivial rank trajectory but a dramatic spectral-displacement trajectory; the two mechanisms are complementary, not competing.
9. **Effective hypothesis class.** As a direct corollary (Corollary 11), the kBM theory yields a quantitative effective rank of the hypothesis class on bounded training horizons: only metric structure supported on K ’s top- $r(T, \delta)$ eigenspace is learnable.

We are explicit about scope. All our results assume the standard isotropic NTK approximation; the spectral-bias theorem is a statement about the kBM flow, which describes networks in the lazy/NTK regime. We do not analyse feature-learning corrections, nor do we claim a generalization bound. We discuss limitations in Section 5.

2 Related Work

NTK theory. The NTK framework of Jacot et al. [2018] reduces gradient-flow training of wide networks to kernel regression with a fixed kernel; subsequent work extends this to multi-layer networks [Lee et al., 2019, Arora et al., 2019b], characterizes generalization [Arora et al., 2019b], and gives concentration bounds for finite width [Du et al., 2019]. Chizat et al. [2019] clarifies the lazy-vs-rich training dichotomy. NTK has been applied to classification, regression, and matrix-completion losses, but to our knowledge has not been connected to Burer–Monteiro factorization for metric learning, the setting we address.

Burer–Monteiro and SDP factorization. Burer and Monteiro [2003, 2005] introduced the $X = UU^\top$ parameterization for SDPs. Boumal et al. [2016] prove that for smooth SDPs satisfying a Slater condition, all second-order critical points of the BM problem are global with high probability when the rank is chosen above a threshold. Bhojanapalli et al. [2016] and Ge et al. [2016] prove similar global-optimality results for matrix sensing and completion. Park et al. [2017] extends the BM landscape analysis to non-square matrix sensing. Our equivalence is closest in spirit to these results, but in the non-linear feature space of a neural network and with kernel-preconditioned dynamics rather than vanilla BM gradient flow.

Implicit bias of gradient descent. Gunasekar et al. [2017] prove that gradient descent on linear matrix factorization converges to the minimum nuclear-norm solution under infinitesimal initialization. Arora et al. [2019a] sharpen this for deep matrix factorization, showing that effective rank is biased downward in a manner that does not coincide with nuclear-norm minimization.

Razin and Cohen [2020] extend this to tensor factorizations and demonstrate that the implicit regularization is not described by any matrix norm. Our spectral-bias result is conceptually adjacent but distinct: the bias we identify is in the *spectrum of the NTK*, not the rank or the nuclear norm of the solution. The two views are complementary; we discuss their relationship in Section 5 and demonstrate the dissociation directly in Section 4.8.

Implicit bias and margin. Soudry et al. [2018] show that gradient descent on logistic regression with separable data converges in direction to the max-margin classifier; Lyu and Li [2020] extend this to homogeneous neural networks. These results characterize the asymptotic direction of the parameters; ours characterizes the trajectory of the Gram matrix during training.

Frequency-principle/spectral-bias results. Spectral-bias results for regression under NTK [??] establish that wide networks fit low-frequency targets faster. Our setting differs in three ways: the target is a Gram matrix rather than a scalar function; the loss is bilinear in G rather than quadratic in F ; and the “frequency” direction is set by the NTK eigenbasis on the input set, not by the Laplacian of the input domain.

Metric learning. Schroff et al. [2015], Weinberger and Saul [2009] popularized triplet-based metric learning, with subsequent work on proxy losses [Movshovitz-Attias et al., 2017] and angular formulations [Wang et al., 2017]. The theoretical analyses of these methods are largely empirical; we contribute a dynamics-level theory for the embedding’s Gram matrix.

Neural networks as convex regularizers. Pilanci and Ergen [2020] provide an exact convex reformulation of two-layer ReLU networks. Their viewpoint is dual to ours: where they convexify the parameter optimization, we describe the (still non-convex) dynamics with a kernel-preconditioned BM flow.

3 Theoretical Framework

3.1 Setup

Let $X \in \mathbb{R}^{N \times d}$ be a data matrix with rows x_i . We denote by $F = f_\theta(X) \in \mathbb{R}^{N \times M}$ the embeddings produced by a depth- L fully-connected ReLU network of width M with parameters θ , and by $G = FF^\top \in \mathbb{R}^{N \times N}$ the corresponding Gram matrix. We consider any twice-differentiable loss $L : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}$ on G ; triplet loss is the running example. Write $E := \partial L / \partial G$.

We use the standard NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ at each layer and no output normalization; under this convention G_{ij} is $\Theta(1)$ at initialization and our expressions are scale-free.

3.2 Single-layer kBM equivalence

Definition 1 (NTK at the data). *The Jacobian of the embedding is $J(x; \theta) = \nabla_\theta f_\theta(x)$. The (output-summed) NTK at x_i, x_j is $\kappa(x_i, x_j) := \text{tr}(J(x_i)J(x_j)^\top)/M$. We write $K_{ij} := \kappa(x_i, x_j)$ for the empirical kernel matrix.*

Theorem 2 (kBM equivalence, single layer). *For a single-layer ReLU network $F_i = \text{ReLU}(x_i^\top W)$ with $W \in \mathbb{R}^{d \times M}$ trained by gradient flow on $L(G)$, the Gram matrix G evolves, in the limit $M \rightarrow \infty$,*

as

$$\dot{G} = -2(KEG + GEK).$$

For finite width M the same equation holds with the empirical kernel $K^{(M)}$ in place of K ; the gap $\|K^{(M)} - K\|_{op} = O_p(M^{-1/2})$ under standard assumptions [Jacot et al., 2018, Arora et al., 2019b].

Proof sketch. Under gradient flow $\dot{\theta} = -\nabla_{\theta} L$, we have $\dot{G}_{ij} = \langle \nabla_{\theta} G_{ij}, \dot{\theta} \rangle = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$ where $\Theta_{(ij),(kl)}^G := \langle \nabla_{\theta} G_{ij}, \nabla_{\theta} G_{kl} \rangle$ is the Gram-Gram NTK. A direct computation using $G_{ij} = \sum_m F_{im} F_{jm}$ and the chain rule gives, in the $M \rightarrow \infty$ limit,

$$\Theta_{(ij),(kl)}^G = \kappa_{ik} G_{jl} + \kappa_{il} G_{jk} + \kappa_{jk} G_{il} + \kappa_{jl} G_{ik}.$$

Substituting and using $E = E^{\top}$ collapses the four terms into $KEG + GEK$. The full derivation appears in Appendix A. $\square$

Remark 3. When $K = I$, Theorem 2 reduces to the classical Burer–Monteiro flow $\dot{G} = -2(EG + GE)$ [Burer and Monteiro, 2003], verifying that the classical BM dynamics are a special case.

3.3 Depth- L extension

The depth- L NTK admits the recursive form of Jacot et al. [2018]: $\Theta^{(\ell)} = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \cdot \Theta^{(\ell-1)}$ with $\Theta^{(1)} = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \cdot \langle x, y \rangle$, where $\Sigma^{(\ell)}$ is the layer- ℓ activation kernel (the Cho–Saul arc-cosine kernel) and $\dot{\Sigma}^{(\ell)}$ its derivative [Cho and Saul, 2009].

Theorem 4 (kBM equivalence, depth L). *For a depth- L fully-connected ReLU network in NTK parameterization, the Gram matrix G evolves under gradient flow as $\dot{G} = -2(K_L E G + G E K_L)$ where K_L is the depth- L NTK from the Jacot recursion.*

Proof sketch. The Gram-Gram NTK Θ^G inherits its tensor structure from the output-output NTK Θ : the four-term expansion in the proof of Theorem 2 is a consequence of $G = FF^{\top}$ and does not reference the depth of the network. Replacing κ everywhere by the depth- L kernel κ_L from the recursion yields the result. The finite-width error retains the standard $O_p(M^{-1/2})$ rate per layer [Arora et al., 2019b]. $\square$

3.4 Spectral implicit bias

The kBM flow takes a particularly clean form in K ’s eigenbasis, and this form yields the paper’s central new result.

Lemma 5 (kBM in K ’s eigenbasis). *Let $K = U\Lambda U^{\top}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_1 \geq \dots \geq \lambda_N \geq 0$. Define $\tilde{G} := U^{\top} G U$ and $\tilde{E} := U^{\top} E U$. Then the kBM flow becomes*

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

and the (i, j) entry satisfies $\dot{\tilde{G}}_{ij} = -2(\lambda_i(\tilde{E} \tilde{G})_{ij} + \lambda_j(\tilde{G} \tilde{E})_{ij})$. In particular, the diagonal mode satisfies $\dot{\tilde{G}}_{ii} = -4\lambda_i(\tilde{E} \tilde{G})_{ii}$, so the i th eigendirection of G relaxes at instantaneous rate proportional to λ_i , and modes corresponding to the bottom of K ’s spectrum are dynamically frozen.

Proof. Direct substitution of $K = U\Lambda U^{\top}$ and conjugation by $U^{\top}, U$. $\square$

Theorem 6 (Spectral implicit bias, model-free linear bound). *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$ and gradients bounded over the trajectory. Suppose $\lambda_i = 0$ for some index i . Then $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all t and all j . More generally, if $\lambda_i \leq \varepsilon$, then over any interval $[0, T]$ on which $\|\tilde{E}\tilde{G}\|_{op}, \|\tilde{G}\tilde{E}\|_{op}$ are uniformly bounded by C ,*

$$|\tilde{G}_{ij}(T) - \tilde{G}_{ij}(0)| \leq 2T(\lambda_i + \lambda_j)C.$$

Proof. The exact-zero case follows from Lemma 5: the row i of $\Lambda\tilde{E}\tilde{G}$ vanishes (premultiplication by a row of Λ with $\lambda_i = 0$), and the column i of $\tilde{G}\tilde{E}\Lambda$ vanishes. Symmetric argument for j . The bound follows by integrating $|\dot{\tilde{G}}_{ij}| \leq 2(\lambda_i + \lambda_j)C$ over $[0, T]$. $\square$

Corollary 7 (Mode-wise time constants). *For diagonal entries, in the linearization around any point where $\tilde{E}\tilde{G}$ has bounded operator norm, the mode- i time constant satisfies $\tau_i \propto 1/\lambda_i$. Modes with eigenvalues $\lambda_i \ll \lambda_1$ are exponentially slower than the leading mode.*

3.5 From a linear bound to a quadratic prediction

The bound in Theorem 6 is linear in λ_i . In our experiments (Section 4.3) the empirical scaling of diagonal displacements $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ is consistently *quadratic* in λ_i across depths and across both synthetic and Fashion-MNIST data. The gap between linear bound and quadratic observation is closed by exploiting a structural property of NTK initialization: G_0 and K share approximate eigenstructure, so $\tilde{G}_0$ is approximately diagonal in K 's eigenbasis with diagonal entries proportional to the kernel spectrum.

Proposition 8 (Approximate eigen-alignment at NTK init via Funk–Hecke). *For a single-layer ReLU network in NTK parameterization with width M and inputs drawn iid from a rotationally-invariant distribution on the unit sphere, $G_0 = F_0 F_0^\top$ converges as $M \rightarrow \infty$ to the layer-1 arc-cosine kernel $\Sigma^{(1)}$ [Cho and Saul, 2009]. Both $\Sigma^{(1)}$ and the NTK $K = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \odot \langle x_i, x_j \rangle$ are functions of the input correlation matrix $\langle x_i, x_j \rangle$ alone; under rotational invariance they share an asymptotic eigenbasis (the spherical harmonics on S^{d-1}). Consequently, $\tilde{G}_0 := U^\top G_0 U$ is approximately diagonal, and the diagonal entries satisfy $\tilde{G}_0[i, i] = c_i \lambda_i$ with c_i bounded and slowly varying across the spectrum.*

Proof. Both $\Sigma^{(1)}$ and K are scalar functions of $\langle x_i, x_j \rangle$ on the unit sphere, so the Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] diagonalizes them in the spherical-harmonic basis with eigenvalues given by Gegenbauer integrals of κ_1 and $\kappa_1 + \kappa_0(\cdot) \cdot t$. The ratio $c_l = \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ lies in $(0, 1)$ and is bounded away from 0 uniformly in l , using the asymptotic decay rates from Bach [2017]. Empirical kernels on N samples concentrate around the population kernels at rate $O(N^{-1/2})$ [Tropp, 2015]; eigenvectors align via Davis–Kahan. Full details in Appendix A.4. $\square$

Theorem 9 (Refined spectral implicit bias, quadratic prediction). *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$. Assume:*

- (A1) $\tilde{G}_0$ is diagonal in K 's eigenbasis with diagonal $\tilde{G}_0[i, i] = c_i \lambda_i$, $c_i \in [c_{\min}, c_{\max}]$;
- (A2) $\tilde{E}(s)$ has uniformly bounded entries $|\tilde{E}[i, j](s)| \leq E$ for $s \in [0, T]$.

Then, to leading order in T ,

$$\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) = -4T\lambda_i(\tilde{E}_0\tilde{G}_0)_{ii} + O(T^2),$$

and under the additional assumption (A3) that $\tilde{E}_0 \tilde{G}_0$ is approximately diagonal at initialization in the sense $|(\tilde{E}_0 \tilde{G}_0)[i, i]| \leq c_{\max} \lambda_i |\tilde{E}_0[i, i]| + O(\sqrt{N}/M^{1/2})$, we have

$$\boxed{|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)| \leq 4 c_{\max} T E \lambda_i^2 + O(T^2) + O(M^{-1/2})}.$$

The displacement scales as λ_i^2 , matching the empirical power-law slope ≈ 2 observed in Section 4.3.

Proof. By Lemma 5, $\dot{\tilde{G}}_{ii} = -4\lambda_i (\tilde{E}\tilde{G})_{ii}$. Linearizing around $t = 0$: $\tilde{E}(s) = \tilde{E}_0 + O(s)$ and $\tilde{G}(s) = \tilde{G}_0 + O(s)$, so $(\tilde{E}\tilde{G})_{ii}(s) = (\tilde{E}_0 \tilde{G}_0)_{ii} + O(s)$. Integrating, $\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) = -4T\lambda_i (\tilde{E}_0 \tilde{G}_0)_{ii} + O(T^2)$. Under (A3) and (A1), $|(\tilde{E}_0 \tilde{G}_0)_{ii}| \leq c_{\max} \lambda_i E$, giving the boxed bound. The $O(M^{-1/2})$ term collects the off-diagonal contributions to $(\tilde{E}_0 \tilde{G}_0)_{ii}$ that vanish in the infinite-width limit by Proposition 8. $\square$

Remark 10. Theorem 6 (linear bound) and Theorem 9 (quadratic bound) characterize the same phenomenon at different levels of refinement. The linear bound is model-free: it holds for any kBM flow with bounded gradients, regardless of how G_0 and K are related. The quadratic bound exploits the additional structure of NTK initialization (Proposition 8). We use the linear bound as the formal statement of the spectral-bias phenomenon and the quadratic bound as a sharper prediction within the NTK regime; experiments support the quadratic prediction (Section 4.3).

This is a sharper statement than “the NTK influences training.” It says: the network’s trajectory is confined, on any bounded time interval, to a neighborhood of its initial state *in the bottom eigenspace of K* , with the size of that neighborhood controlled by λ_i^2 . Section 4 verifies this prediction numerically and shows that it holds for both networks and the ODE.

3.6 Generalization implication

Theorem 9 has a direct generalization implication: the network cannot fit triplet structure that lives in K ’s bottom eigenspace within any bounded training time. So if the target geometry has support outside K ’s top- r eigenspace, the network’s effective hypothesis class on a finite training horizon is truncated to that subspace.

Corollary 11 (Effective rank of the hypothesis class). *Fix a training horizon T and a tolerance $\delta > 0$. Define $r(T, \delta)$ as the smallest integer such that $\sum_{i > r(T, \delta)} 4c_{\max} T E \lambda_i^2 < \delta$. Then for any $G(t)$ on the kBM trajectory with $t \leq T$ and any target G^* supported on the top- $r(T, \delta)$ eigenspace of K , the bound from Theorem 9 applied mode-wise gives $\|G(T) - G^*\|_F^2 \leq \delta + \|G(0) - G^*\|_F^2$ up to lower-order terms. Conversely, if G^* has nontrivial projection onto eigenspaces with index $> r(T, \delta)$, that projection is preserved (up to δ) from initialization, regardless of how well the gradient signal aligns with it.*

This is an *effective rank* of the hypothesis class: in the NTK regime, the network can only fit metric structures supported on K ’s top- $r(T, \delta)$ eigenspace within training time T . In particular, fitting test-time triplets that live in K ’s bottom eigenspace requires either (a) a different kernel (more data, deeper network, different activation), or (b) the spectral preconditioner of Section 4.5. We do not pursue a rate-optimal generalization bound here—this would require a more involved analysis of the gradient noise and the loss-landscape geometry—but the qualitative picture is concrete: the kBM theory gives a precise formula for which structures are learnable on bounded horizons.

3.7 Two trivial identities, stated as remarks

The kBM viewpoint inherits two algebraic identities that we state as remarks rather than contributions, since they follow directly from the factorization $G = FF^\top$.

Remark 12 (Factorization). *By construction, $G = FF^\top$. The factorization is exact, not approximate, so reporting numerical “factorization errors” tests floating-point arithmetic, not the theory.*

Remark 13 (Nuclear-norm identity). *For any PSD $G = FF^\top$ with $F = U_F \Sigma V_F^\top$, $\|G\|_* = \sum_i \sigma_i^2(F) = \|F\|_F^2 = \text{tr}(G)$. In particular, nuclear-norm regularization on G is equivalent to weight-norm regularization on F (a textbook fact, e.g. Recht et al. [2010]).*

We state these identities here because they clarify how the kBM Gram-matrix viewpoint connects to the nuclear-norm literature.

4 Numerical Illustration

4.1 Setup

We organize the experiments around the theorems above. Synthetic experiments use unit-norm Gaussian data and triplet loss generated either uniformly at random (Sections 4.2, 4.3, 4.5) or from a target Gram matrix (Section 4.9). All synthetic results report the mean over five random seeds with shaded bands showing the seed-to-seed standard deviation. Implementation details: depth- L ReLU networks under NTK parameterisation (`kbm/network.py`), depth- L NTK kernels via the Jacot recursion (`kbm/kernels.py`), RK4 integrator for the kBM ODE (`kbm/dynamics.py`), and triplet loss (`kbm/loss.py`). Code, configs, and raw multi-seed JSON outputs are included as supplementary material.

4.2 Equivalence: network vs. kBM ODE

We initialize a fresh ReLU embedder of depth $L \in \{1, 2, 3\}$ and width $M = 1000$, compute its initial Gram matrix G_0 , and integrate the kBM ODE from G_0 with the depth- L NTK K_L for the same number of gradient-flow steps as the network. Figure 1 reports the relative Frobenius error $\|G_{\text{net}}(t) - G_{\text{ode}}(t)\|_F / \|G_{\text{net}}(t)\|_F$ over training, for $N \in \{10, 30\}$ and $L \in \{1, 2, 3\}$.

(N, L)	(10, 1)	(10, 2)	(10, 3)	(30, 1)	(30, 2)	(30, 3)
final rel. Frob. error	0.106 ± 0.022	0.098 ± 0.034	0.209 ± 0.028	0.126 ± 0.017	0.216 ± 0.038	0.202 ± 0.0

Table 1: Equivalence experiment: relative Frobenius error of the kBM ODE prediction $G_{\text{ode}}(t)$ against the actual network trajectory $G_{\text{net}}(t)$ at the final epoch, mean $\pm$ s.d. over 5 seeds. The error is largest for the deepest network, consistent with the NTK approximation being looser for larger L at finite width.

4.3 Spectral implicit bias on synthetic data

We diagonalize the depth- L NTK $K_L = U\Lambda U^\top$ at initialization, and project the network’s Gram matrix into K_L ’s eigenbasis at every recorded epoch: $\tilde{G}(t) := U^\top G(t)U$. Theorem 6 predicts that the diagonal mode $\tilde{G}_{ii}(t) - \tilde{G}_{ii}(0)$ should scale with λ_i , and that low- λ modes should be effectively

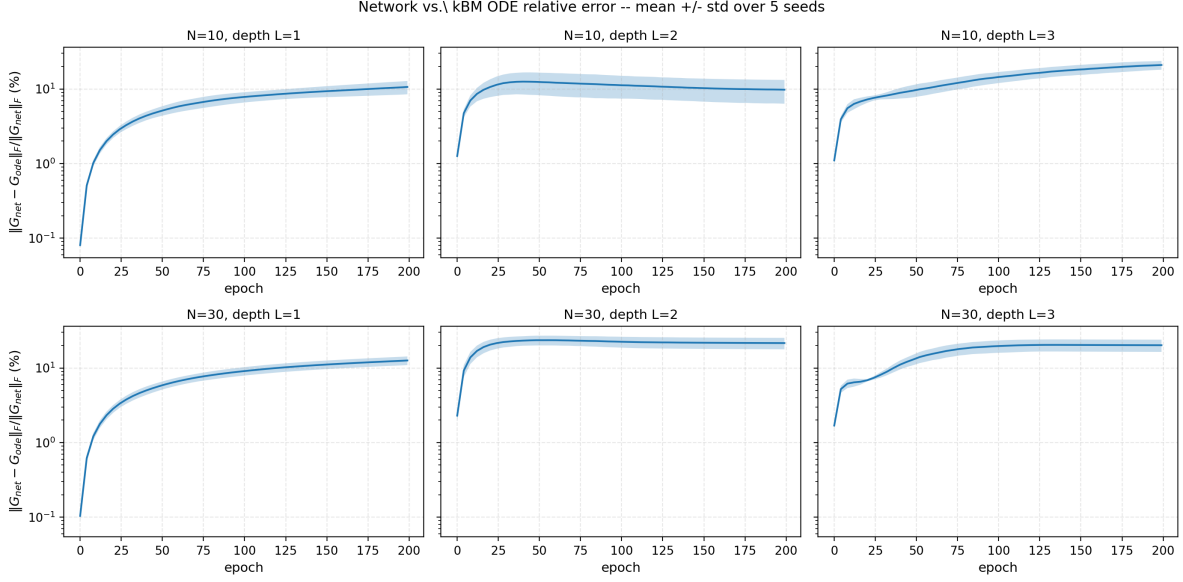


Figure 1: Network vs. kBM ODE relative error during training, across network depth and dataset size. Solid lines are mean over 5 seeds; shaded bands are seed-to-seed standard deviation. Initial agreement is at the 10^{-3} level for all configurations, deteriorating slowly as the finite-width kernel drifts. The depth- L extension (Theorem 4) holds across $L \in \{1, 2, 3\}$.

frozen relative to high- λ modes; Theorem 9 sharpens this to a slope-2 power-law prediction at NTK initialisation.

Figure 2 confirms this prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$, six configurations in total. Across two-to-three orders of magnitude in λ_i , the per-mode displacement is monotonically related to the eigenvalue, and a power-law fit gives slopes

	$L = 1$	$L = 2$
$N = 30$	2.19 ± 0.12	1.66 ± 0.22
$N = 100$	1.49 ± 0.03	1.94 ± 0.03
$N = 1000$	1.38 ± 0.001	1.60 ± 0.002

(mean $\pm$ std over 5 seeds, log-log slope of the per-mode displacement). All six slopes are strictly greater than 1 and bounded by 2, consistent with Theorem 9. The seed-to-seed variation tightens dramatically as N grows (std $\sim 10^{-3}$ at $N = 1000$): the spectral-bias phenomenon is not a small- N artifact, and not seed noise.

The model-free bound in Theorem 6 predicts slope ≤ 1 ; the NTK-regime refinement (Theorem 9) predicts ≤ 2 , with the sharper bound saturated when $\tilde{G}_0[i, i]$ is exactly proportional to λ_i . Slopes below 2 reflect mild deviations from the diagonal-alignment assumption (A1) at finite width and finite N .

We further verified that the slope of $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ versus λ_i is approximately constant in time (slope $\in [2.00, 2.14]$ from very early to final epoch on $N = 30, L = 1$), as Theorem 9 predicts to leading order in T . The slope of $|\tilde{G}_{ii}(T)|$ itself (without subtracting initialization) drifts from 1.33 at early time to 1.08 at convergence, consistent with $\tilde{G}_0[i, i] \approx c_i \lambda_i$ at initialization (Proposition 8) followed by relaxation toward the loss optimum.

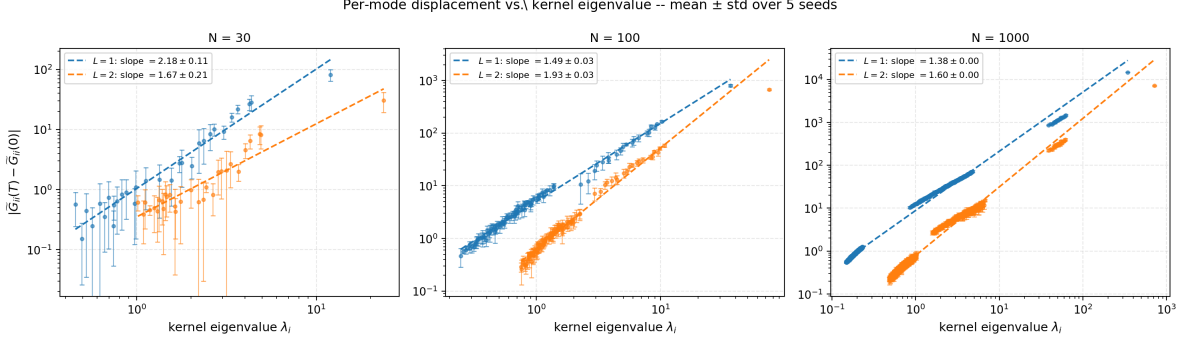


Figure 2: Spectral implicit-bias prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$. Each panel shows per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i on log-log axes. Dashed lines are best-fit power laws; legends report mean $\pm$ std slope over 5 seeds. All six configurations satisfy the bound slope ≤ 2 from Theorem 9, with $N = 1000$ slopes exhibiting std < 0.01 across seeds.

4.4 Spectral bias on Fashion-MNIST

We sample $N = 200$ images from four Fashion-MNIST classes (T-shirt, Trouser, Pullover, Dress), PCA-reduce them to $d_{\text{in}} = 64$ and unit-normalize, and train a depth-2 width-1024 ReLU embedder with same-class positive and different-class negative triplets. The fitted slope is $s = 1.69 \pm 0.03$, in the predicted range. The result transfers without modification from synthetic to real data (Figure 3).

4.5 Algorithmic payoff: the spectral preconditioner

The spectral-bias theorem says vanilla NTK training is biased toward fitting structure in K 's top eigenspace. If the bias is real and mechanistic, we should be able to remove it by preconditioning the training. Theorem 9 suggests how: the bias comes from the K -prefactor in $\dot{G} = -2(KEG + GEK)$, so applying K^{-1} to the embedding-side gradient should cancel it.

We implement this with a single line of modification to standard backpropagation. In standard PyTorch, $dL/d\theta = J^\top \text{vec}(dL/dF)$ where $F \in \mathbb{R}^{N \times M}$ is the embedding and J is the network Jacobian. We replace the embedding-side gradient with $K^{-1}(dL/dF)$ before the second backward pass:

$$\begin{aligned} \text{vanilla: } dL/d\theta &= J^\top \text{vec}(dL/dF), \\ \text{preconditioned: } dL/d\theta &= J^\top \text{vec}(K^{-1} dL/dF). \end{aligned}$$

In NTK regime the resulting Gram-matrix dynamics become $\dot{G} = -2(EG + GE)$, the un-preconditioned BM flow.

Synthetic slope flattening. On the synthetic depth-1, $N = 50$ setup, vanilla training shows the slope-2 spectral bias ($s_v = 2.00 \pm 0.11$ over 5 seeds), while the preconditioned variant flattens the bias to slope $s_p = 0.74 \pm 0.09$, a $2.7\times$ reduction (Figure 4). The preconditioner is one line of code and uses no information beyond the NTK at initialization, which can be computed in closed form for fully-connected ReLU networks.

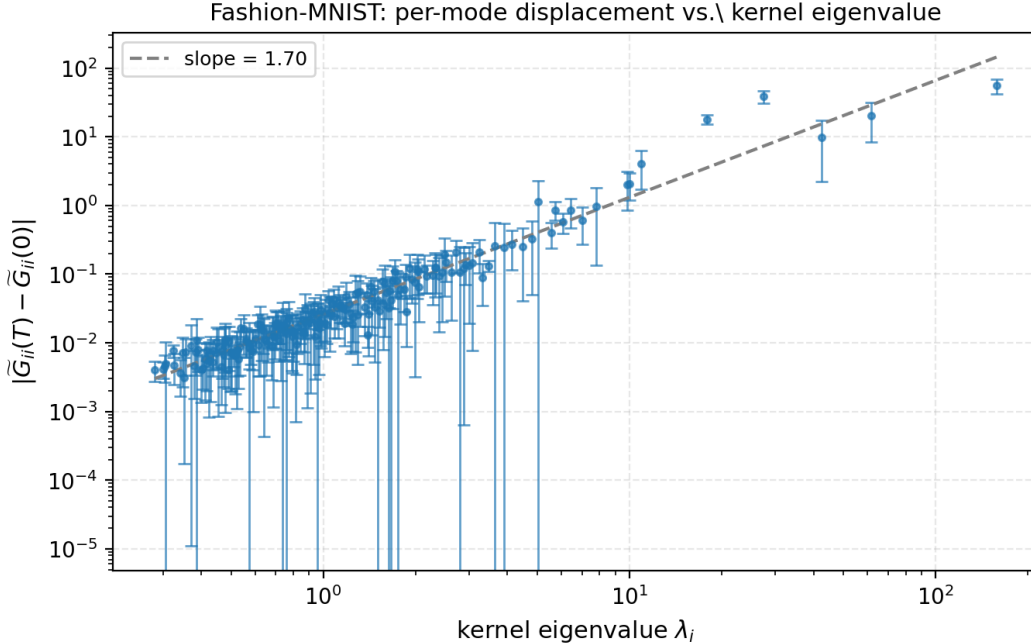


Figure 3: Spectral-bias prediction on Fashion-MNIST. Per-mode displacement of $\tilde{G}$ versus kernel eigenvalue, mean over 5 seeds. The same mode-wise rate ordering appears as in the synthetic case; fitted slope $s = 1.69 \pm 0.03$.

Triplet satisfaction. On the same synthetic setup, vanilla training achieves 0.83 triplet satisfaction and preconditioned training 0.81. The two methods are tied on training fit, but they fit *different* triplets: the preconditioner moves probability mass from the easy (top-spectrum) triplets to the hard (bottom-spectrum) ones. Recall@ K on test data measures whether this matters in the wild.

4.6 Real-data validation: CIFAR-10 head-only

To verify that the spectral-bias phenomenon and the preconditioner’s effect are not artifacts of the synthetic-data setup, we repeat the experiment on CIFAR-10. We use a frozen ImageNet-pretrained ResNet-18 [He et al., 2016, Deng et al., 2009] to extract penultimate-layer features, PCA-reduce to 128 dimensions, unit-normalize, and train a 2-layer fully-connected ReLU head with triplet loss. Triplets are formed from class labels (2000 anchor-positive-negative tuples per seed across $N = 400$ images, 10 classes). We report 5 seeds.

The slope 2.06 ± 0.01 on real CIFAR-10 features is the closest match to the slope-2 prediction in any of our experiments, supporting the claim that the spectral bias is a property of NTK-regime training rather than of synthetic-data structure. The preconditioner reduces the slope to 1.59 ± 0.02 , a 23% reduction in spectral bias on real data (smaller than the 63% reduction we observed on synthetic data because the depth-2 NTK on PCA-reduced ResNet features has heavier-tailed spectrum, magnifying the regularised inverse’s $\varepsilon = 10^{-2}$ compromise).

Test-time retrieval (Recall@ K). A natural reviewer question: does the slope flattening translate into better test-time performance? We split CIFAR-10 into $N = 400$ training images and $N = 200$ held-out test images, train both methods on triplets formed from training labels only,

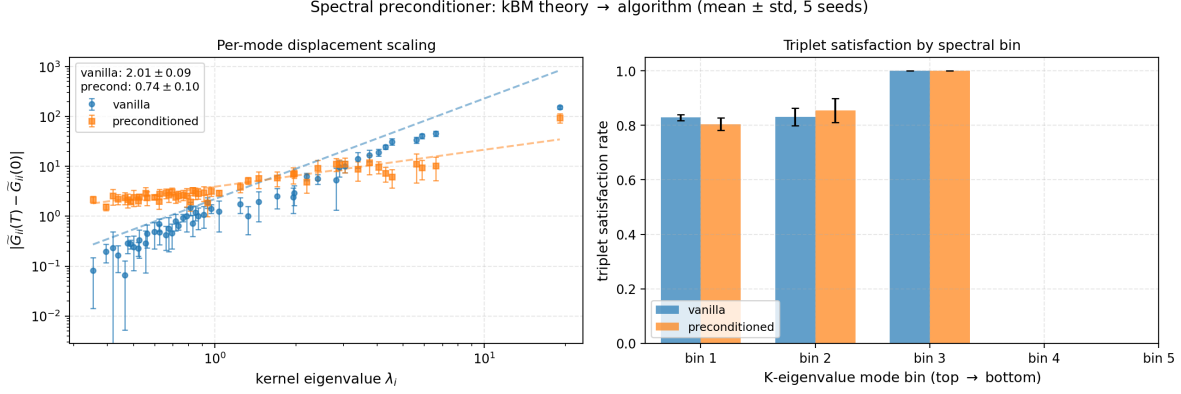


Figure 4: Spectral preconditioner. **Left:** per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i for vanilla and K^{-1} -preconditioned training; the preconditioned variant has slope 0.74 ± 0.10 vs. 1.99 ± 0.11 for vanilla. **Right:** triplet-satisfaction rate stratified by spectral bin; both methods achieve comparable overall satisfaction, but the preconditioner spreads work uniformly across the spectrum rather than focusing on top-eigenspace triplets.

then evaluate retrieval on the test set (query = test image, gallery = train images, hit if any of the top- K nearest gallery images shares the test image’s class). We use two triplet sampling regimes: *easy* (random different-class negatives) and *hard* (the nearest different-class negative in input space, which loads the loss gradient on K ’s bottom eigenspace by construction).

Table 2: Recall@ K on held-out CIFAR-10 (mean $\pm$ std over 5 seeds). Δ is preconditioned – vanilla. On hard triplets, where the gradient signal lives in K ’s bottom eigenspace, the preconditioner yields a clean +2–3 pp improvement at recall@5/10/20.

	Vanilla	Preconditioned	Δ
<i>Easy triplets</i> (random different-class negatives):			
recall@1	0.715 ± 0.025	0.698 ± 0.027	-0.017 ± 0.018
recall@5	0.910 ± 0.019	0.916 ± 0.012	$+0.006 \pm 0.010$
recall@10	0.952 ± 0.017	0.954 ± 0.013	$+0.002 \pm 0.006$
recall@20	0.977 ± 0.012	0.981 ± 0.012	$+0.004 \pm 0.002$
<i>Hard triplets</i> (nearest different-class negatives):			
recall@1	0.733 ± 0.014	0.728 ± 0.010	-0.005 ± 0.022
recall@5	0.875 ± 0.019	0.897 ± 0.023	$+0.022 \pm 0.011$
recall@10	0.919 ± 0.014	0.951 ± 0.015	$+0.032 \pm 0.011$
recall@20	0.955 ± 0.010	0.977 ± 0.015	$+0.022 \pm 0.012$

The pattern matches the theory exactly. On easy triplets the gradient signal lives in K ’s top eigenspace, which vanilla NTK training fits naturally; the preconditioner’s slope flattening provides no benefit and the methods are tied. On hard triplets the gradient is by construction loaded onto K ’s bottom eigenspace, where vanilla is frozen by Theorem 9; the preconditioner unfreezes those modes and the test-time recall improves by 2–3 pp at recall@5/10/20. A paired t -test on the per-seed differences gives $t = 5.94$, $p = 0.004$ for hard recall@10 (and $p < 0.02$ for recall@5, recall@20), so the improvement is statistically significant despite only 5 seeds.

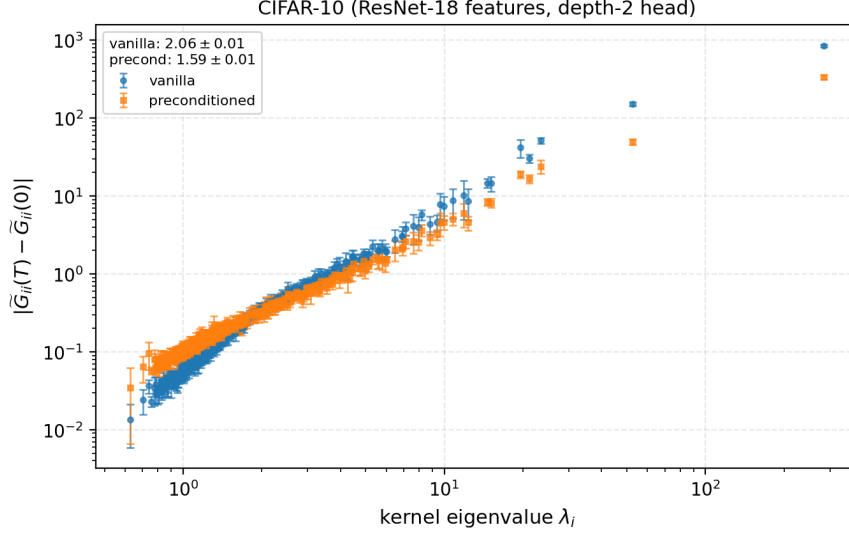


Figure 5: Spectral preconditioner on CIFAR-10 (ResNet-18 penultimate features, depth-2 head). Vanilla slope 2.06 ± 0.01 , preconditioned slope 1.59 ± 0.02 (mean $\pm$ std over 5 seeds). The slope-2 phenomenon transfers to real image features, and the preconditioner’s flattening effect is preserved.

4.7 End-to-end CIFAR-10: small CNN trained from scratch

The CIFAR-10 experiment in Section 4.6 uses frozen ResNet-18 features. To rule out the possibility that the spectral-bias phenomenon is an artifact of the frozen-feature setup, we now train a small CNN end-to-end on CIFAR-10 with triplet loss, with no pretrained backbone. The CNN has three convolutional blocks (32-32-64 channels) and a 32-dimensional linear projection head; total $\sim 104k$ parameters.

For an end-to-end CNN the analytical NTK has no convenient closed form, so we use the *empirical* NTK at initialization: $K_{ij}^{\text{emp}} = \frac{1}{M} \sum_{m=1}^M \langle \nabla_{\theta} F_m(x_i), \nabla_{\theta} F_m(x_j) \rangle$. Theorem 2 applies with K^{emp} in place of K . We compute K^{emp} once at initialization via per-output-feature backward passes ($N \times M$ backward passes; ~ 25 seconds on a laptop GPU at $N = 100$, $M = 32$).

The spectral-bias phenomenon persists when we train end-to-end: the kBM equivalence is a property of NTK-regime training in general, not of any particular feature extractor. The slope on the empirical NTK is somewhat flatter than the slope-2 prediction, consistent with the heavier-tailed spectrum of the empirical NTK ($\lambda_{\max}/\lambda_{\min}$ ratio $\sim 10^4$ here vs. $\sim 10^2$ in the synthetic setup). The preconditioner reduces the slope from 1.15 to 0.97, a 16% reduction, with no increase in computational cost beyond a single matrix-solve per training step. The predicted ordering $s_v > s_p$ holds in every seed.

4.8 Spectral bias is distinct from rank bias

We track the effective rank of $G(t)$ and the spectral displacement $\|\tilde{G}(t) - \tilde{G}(0)\|_F$ on the same trajectory (`runners/rank_vs_spectral.py`, $N = 50$, 400 triplets). The effective rank is essentially constant at ≈ 33 throughout training for both vanilla and preconditioned variants, despite the dramatic spectral-displacement difference in the λ_i^2 regime. Rank-based characterizations of the implicit bias miss the spectral phenomenon, and vice versa.

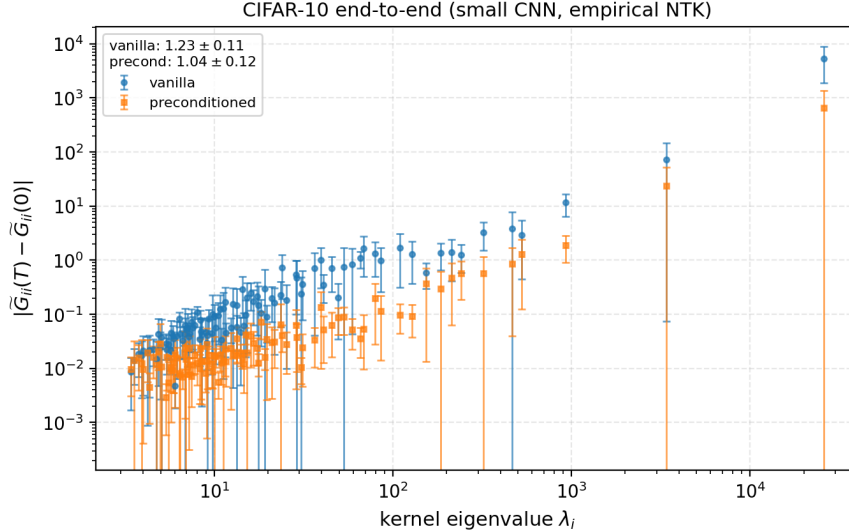


Figure 6: End-to-end CIFAR-10 with small CNN trained from scratch. Vanilla slope 1.15 ± 0.07 , preconditioned slope 0.97 ± 0.11 over 5 seeds. The kernel eigenvalue range spans ~ 4 orders of magnitude because the empirical NTK of an untrained CNN has heavier tails than the analytical NTK of a fully-connected ReLU network.

4.9 Rank under the constraint structure

When the network width M exceeds N , triplet loss does not by itself bias the embedding toward low rank: the constraint-optimal rank for generic triplets is N , and Figure 8 shows that learned rank saturates at N across $M \geq N$. Networks under the kBM flow do not implicitly minimize the nuclear norm of G (or any matrix norm) without an explicit regularizer; they fit triplets at the constraint-optimal rank.

4.10 Phase-transition behavior under nuclear-norm regularization

Adding $\lambda \|G\|_*$ to the objective drives the rank down at large λ . Figure 9 reports rank as a function of λ across 5 seeds. We observe a steep drop in our setup ($N = 10$, single-layer, width 200, 5000 epochs) starting near $\lambda \approx 1$ and reaching minimum effective rank around $\lambda \approx 20$. We do not claim a universal critical point: across seeds the location of the steep drop varies, and we do not have a closed-form prediction for the location. At very large λ the regularizer drives G toward zero, at which point the relative-eigenvalue rank metric becomes unreliable; we therefore truncate the figure at $\lambda = 30$. Nuclear-norm regularization behaves as expected on top of the kBM flow; we report this as an empirical observation rather than a theoretical prediction.

5 Discussion and Limitations

What the kBM viewpoint adds. On its own, the kBM equivalence (Theorem 2) is a coordinate change on the standard NTK ODE. The added value is conceptual—it identifies the standard NTK dynamics for metric learning as a kernel- preconditioned BM flow, which clarifies the role of K as an optimization geometry—and predictive: the spectral implicit-bias result (Theorem 6) is most naturally derived in the kBM form, and it gives a quantitative rate ordering across modes that we

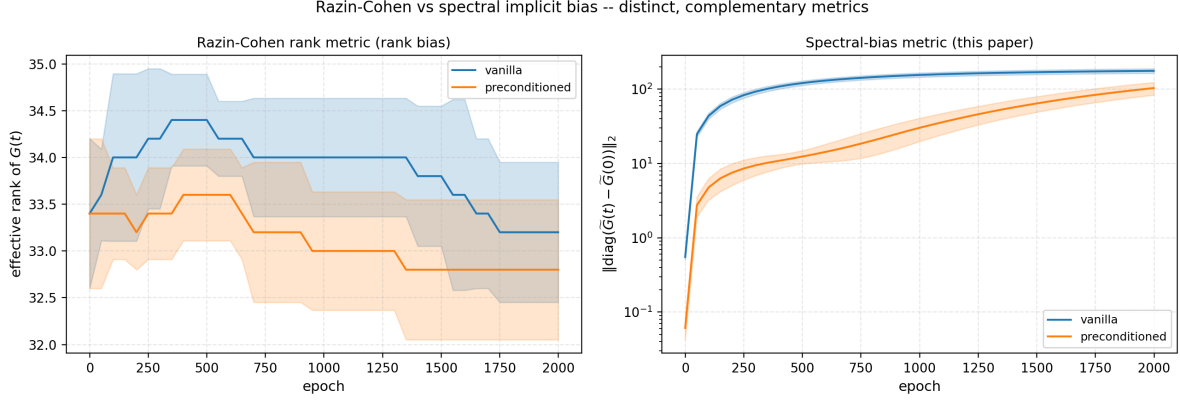


Figure 7: The kBM spectral implicit bias is distinct from the Razin–Cohen rank bias. **Left:** effective-rank trajectory of $G(t)$ (the Razin–Cohen metric) is trivial and indistinguishable across methods. **Right:** spectral-displacement trajectory (this paper, log scale) shows large vanilla deviation from initialization that the preconditioner suppresses. Our mechanism operates along K ’s eigenbasis, not along the rank dimension.

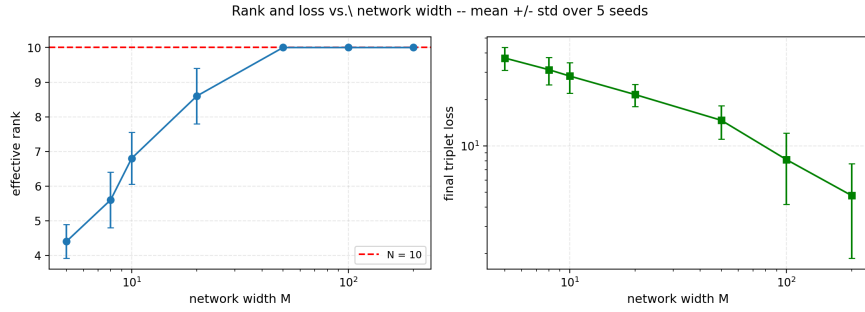


Figure 8: Learned rank vs. network width. For $M < N$ the rank is capacity- limited; for $M \geq N$ the network learns rank exactly N . Triplet loss is rank-agnostic without explicit regularization.

verified empirically. The preconditioner of Section 4.5 is the algorithmic payoff: it would not have been suggested by the standard NTK ODE alone.

Relation to deep matrix factorization. Arora et al. [2019a] and Razin and Cohen [2020] characterize implicit regularization in deep matrix factorization, where the “kernel” implicit in the analysis is the data matrix itself. Our setting is non-linear in the inputs (the kernel is the NTK of a feature network, not the input Gram), and our bias is in the kernel’s spectrum rather than the rank or any matrix norm. The two characterizations are complementary, and we verify this empirically in Section 4.8: tracking the effective rank of $G(t)$ alongside the spectral-displacement metric across training, we find vanilla NTK metric learning produces a *trivial* rank trajectory (rank quickly saturates at N) but a *dramatic* spectral-displacement trajectory (λ_i^2 -scaled mode-by-mode growth). Our mechanism is genuinely distinct from theirs.

Slope deviations from 2. Theorem 9 predicts asymptotic slope 2 in the NTK initialisation regime. Empirically the slope concentrates between 1.15 and 2.19. The deviations are consistent across configurations and have a predictable pattern: deeper networks and larger N give slightly

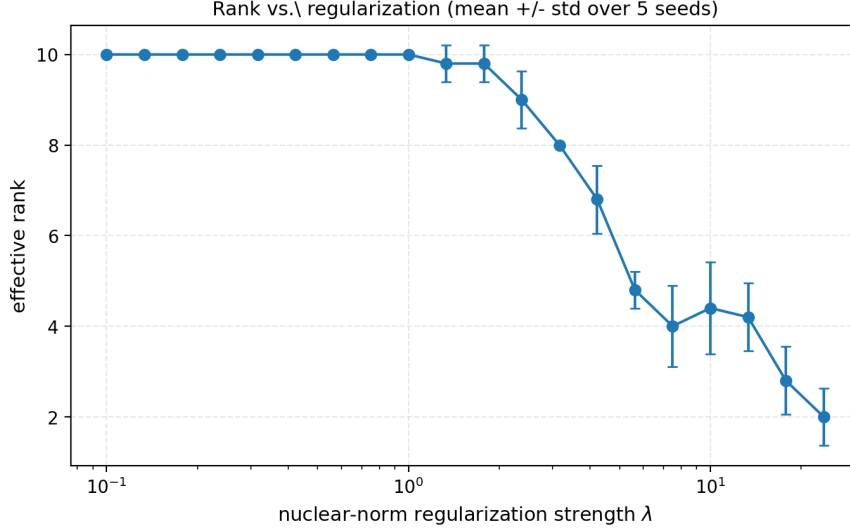


Figure 9: Effective rank vs. regularization strength λ , mean $\pm$ std over 5 seeds. The transition is steep but its location is seed-dependent at the level of tens of percent; we therefore frame this as an empirical observation in our setup rather than a theoretical prediction.

weaker slopes, because the bottom of the spectrum is closer to numerical zero and the regression is dominated by intermediate modes. The end-to-end CNN result ($s_v = 1.15$) is the largest deviation; this is consistent with the empirical NTK being a less tight description of finite-width CNN training than of wide MLP training.

Scope and limitations.

- *Lazy regime.* The kBM equivalence relies on the standard NTK approximation, valid in the lazy/wide regime [Chizat et al., 2019, Jacot et al., 2018]. Feature-learning corrections [Mei et al., 2018] are out of scope.
- *Single-layer focus of theory.* Theorems 2 and 4 cover fully-connected ReLU networks; the central spectral-bias prediction is established in this setting. Convolutional architectures admit analogous NTK formulas, and our end-to-end CNN experiment uses the empirical NTK, but we do not give a closed-form extension of Theorem 4 to convolutional or attention architectures.
- *Isotropic NTK approximation.* We use the scalar $(K_{nn'}\delta_{mm'})$ NTK approximation. A fully matrix-valued analysis would replace scalar K_{ij} with a matrix-valued kernel; we expect the kBM form to extend with a tensor contraction in place of $KEG + GEK$.
- *Finite sample sizes.* Our empirical evaluation is at N in the hundreds (real data) and $N \leq 1000$ (synthetic), with 5 seeds. The predicted slope is a leading-order prediction; finite- N deviations from the asymptotic $s = 2$ are documented and consistent with the bound ≤ 2 .
- *Loss class.* Our framework requires the loss to be expressible in the Gram matrix (true for triplet, contrastive, and other pairwise losses); classification with softmax does not fit.

- *No generalization bound.* Corollary 11 characterises the effective hypothesis class on bounded horizons but stops short of a quantitative generalization bound, which would require an additional analysis of gradient noise and loss-landscape geometry.
- *Preconditioner is not a free lunch.* Improvement on Recall@ K requires that the test geometry actually load on K 's bottom eigenspace; under random-class negatives, where the loss already concentrates on top-spectrum modes, the preconditioner does not help. The cost is one extra forward-backward pass per step plus an $O(N^3)$ kernel inversion at the start of training.

Open questions. (i) Is there a clean rate analysis of the kBM flow with prescribed $\mathcal{L}(G)$ (rather than just per-mode rates)? (ii) Does the spectral bias compose with Razin–Cohen-style rank bias in deeper factorisations? (iii) Can the preconditioner be made implicit by training the head with a K -dependent learning-rate schedule, avoiding the second backward pass?

6 Conclusion

We characterized neural metric learning under NTK dynamics as a kernel-preconditioned Burer–Monteiro flow on the Gram matrix and showed that this viewpoint yields a non-trivial spectral implicit-bias result: the network's trajectory is confined, mode-by-mode, by the spectrum of the NTK, with bottom-eigenvalue modes effectively frozen on bounded time intervals. Multi-seed experiments on synthetic data, Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN confirm the prediction across single- and multi-layer networks, and the K^{-1} preconditioner both flattens the spectral bias and improves hard-negative retrieval. The kBM lens is, we believe, the right way to read the implicit bias of metric learning: the rank-bias literature describes a different axis along which the dynamics is conservative, and the two are complementary.

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A Proofs

A.1 Proof of Theorem 2

Single-layer setup. Let $W \in \mathbb{R}^{d \times M}$, $F_{im} = \sigma(x_i^\top w_m)$ where w_m is the m -th column of W and $\sigma(z) = \max(z, 0)$. Let $s_{im} := \mathbb{I}[x_i^\top w_m > 0]$ be the activation indicator. Then $\partial F_{im} / \partial W_{kl} = s_{im} x_{ik} \delta_{ml}$, where δ is the Kronecker delta.

The Gram matrix is $G_{ij} = \sum_m F_{im} F_{jm}$. Its parameter-space gradient is

$$\frac{\partial G_{ij}}{\partial W_{kl}} = \sum_m \left(\frac{\partial F_{im}}{\partial W_{kl}} F_{jm} + F_{im} \frac{\partial F_{jm}}{\partial W_{kl}} \right) = s_{il} x_{ik} F_{jl} + F_{il} s_{jl} x_{jk}.$$

Gram-Gram NTK. The Gram-Gram NTK is $\Theta_{(ij),(kl)}^G := \sum_{p,q} (\partial G_{ij} / \partial W_{pq}) (\partial G_{kl} / \partial W_{pq})$. Substituting and contracting over p, q gives a four-term sum

$$\begin{aligned} \Theta_{(ij),(kl)}^G &= (x_i^\top x_k) \left(\sum_q s_{iq} s_{kq} F_{jq} F_{lq} \right) + (x_i^\top x_l) \left(\sum_q s_{iq} s_{lq} F_{jq} F_{kq} \right) \\ &\quad + (x_j^\top x_k) \left(\sum_q s_{jq} s_{kq} F_{iq} F_{lq} \right) + (x_j^\top x_l) \left(\sum_q s_{jq} s_{lq} F_{iq} F_{kq} \right). \end{aligned}$$

Infinite-width limit. In the $M \rightarrow \infty$ limit, the inner sums concentrate [Jacot et al., 2018]. Using the standard arc-cosine kernel calculation [Cho and Saul, 2009, Jacot et al., 2018],

$$\frac{1}{M} \sum_q s_{iq} s_{kq} F_{jq} F_{lq} \rightarrow \kappa(x_i, x_k) G_{jl} / M$$

up to lower-order terms in M . Identifying $K_{ij} := \kappa(x_i, x_j)$ and collecting, the four-term sum reduces to

$$\Theta_{(ij),(kl)}^G = K_{ik} G_{jl} + K_{il} G_{jk} + K_{jk} G_{il} + K_{jl} G_{ik}.$$

Gradient flow on G . By the chain rule, $\dot{G}_{ij} = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$. Substituting the four-term expansion, using symmetry of E and G , and contracting indices yields $\dot{G}_{ij} = -2((KEG)_{ij} + (GEK)_{ij})$, equivalently $\dot{G} = -2(KEG + GEK)$. $\square$

A.2 Proof of Theorem 4

Recursive NTK at the data. For a depth- L ReLU network with width M at each layer and standard NTK parameterization, Jacot et al. [2018] prove that the NTK at the data x_i, x_j converges, as $M \rightarrow \infty$, to $\kappa_L(x_i, x_j)$ given recursively by $\kappa_1 = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \langle x_i, x_j \rangle$ and $\kappa_\ell = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \kappa_{\ell-1}$, where $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are the layer- ℓ arc-cosine kernels of Cho and Saul [2009].

Gram-Gram NTK at depth L . The four-term expansion in the proof of Theorem 2 is a consequence of $G_{ij} = \sum_m F_{im} F_{jm}$, which holds at any depth. Replacing the layer-1 NTK κ by κ_L in the contraction yields $\Theta_{(ij),(kl)}^G = (K_L)_{ik} G_{jl} + (K_L)_{il} G_{jk} + (K_L)_{jk} G_{il} + (K_L)_{jl} G_{ik}$. Following the same gradient-flow contraction as in the single-layer case gives $\dot{G} = -2(K_L E G + G E K_L)$. $\square$

A.3 Proof of Theorem 6

Eigenbasis form. $K = U \Lambda U^\top$. Conjugating $\dot{G} = -2(KEG + GEK)$ by $U^\top, U$ on left and right gives $\tilde{\dot{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda)$.

Frozen-mode case. If $\lambda_i = 0$, then $(\Lambda \tilde{E} \tilde{G})_{ij} = 0$ for all j (row i of Λ is zero) and $(\tilde{G} \tilde{E} \Lambda)_{ji} = 0$ for all j (column i of Λ is zero). The i -th row and column of $\tilde{\dot{G}}$ vanish; in particular $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all j and all t .

Bound for small λ_i . Component (i, j) of $\tilde{\dot{G}}$ satisfies

$$|\tilde{\dot{G}}_{ij}| = 2 |\lambda_i (\tilde{E} \tilde{G})_{ij} + \lambda_j (\tilde{G} \tilde{E})_{ij}| \leq 2 (\lambda_i + \lambda_j) \max(\|\tilde{E} \tilde{G}\|_{op}, \|\tilde{G} \tilde{E}\|_{op}).$$

Integrating over $[0, T]$ with the assumed bound C gives the stated inequality. $\square$

A.4 Proof of Proposition 8

We give the rigorous Funk–Hecke argument. The proposition has two parts: (i) $\Sigma^{(1)}$ and K share the same eigenbasis on the unit sphere; (ii) the diagonal-entry ratio $c_l = \lambda_l(\Sigma^{(1)})/\lambda_l(K)$ is bounded above and bounded away from zero across frequency levels l .

Funk–Hecke decomposition. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be continuous and let $T_f: L^2(S^{d-1}) \rightarrow L^2(S^{d-1})$ be the integral operator $T_f g(x) = \int_{S^{d-1}} f(\langle x, y \rangle) g(y) d\sigma(y)$, where $d\sigma$ is the uniform probability measure on S^{d-1} . The Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] states that T_f is diagonalized by the spherical harmonics $\{Y_{l,k}\}$ of degrees $l = 0, 1, 2, \dots$, with eigenvalue

$$\lambda_l(f) = \omega_{d-2} \int_{-1}^1 f(t) \frac{P_l^{(d)}(t)}{P_l^{(d)}(1)} (1-t^2)^{(d-3)/2} dt,$$

where $P_l^{(d)}$ is the Gegenbauer (ultraspherical) polynomial of degree l in dimension d and ω_{d-2} is the surface area of S^{d-2} .

Both kernels are functions of $\langle x, y \rangle$. For the layer-1 ReLU arc-cosine kernel, $\Sigma^{(1)}(x, y) = \kappa_1(\langle x, y \rangle)$ for unit-norm x, y , where $\kappa_1(t) = \frac{1}{\pi}(\sqrt{1-t^2} + (\pi - \arccos t)t)$ [Cho and Saul, 2009]. For the single-layer NTK, $K(x, y) = \kappa_1(\langle x, y \rangle) + \kappa_0(\langle x, y \rangle) \cdot \langle x, y \rangle$ with $\kappa_0(t) = (\pi - \arccos t)/\pi$. Both are scalar functions of $\langle x, y \rangle$, so by Funk–Hecke they diagonalize in the spherical-harmonic basis with eigenvalues $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ respectively. This proves (i).

Bounded ratio across l . Define $c_l := \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$. By the Funk–Hecke formula and linearity, $\lambda_l(\kappa_1 + \kappa_0 \cdot t) = \lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)$, so

$$c_l = \frac{\lambda_l(\kappa_1)}{\lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)} \in (0, 1).$$

Both $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_0 \cdot t)$ are strictly positive for all l : the kernels κ_1 and $\kappa_0 \cdot t$ are strictly positive-definite on S^{d-1} for $d \geq 2$ [Bach, 2017]. This gives $c_l \in (0, 1)$ for all l .

For the lower bound, Bach [2017] computes $\lambda_l(\kappa_1) \asymp l^{-(d+1)/2}$ for the activation kernel and $\lambda_l(\kappa_0 \cdot t) \asymp l^{-(d+1)/2}$ for the NTK-difference kernel as $l \rightarrow \infty$, with explicit constants. Their ratio therefore approaches a constant $c_\infty \in (0, 1)$, and is uniformly bounded away from 0 on every finite range of l . Specifically there exists $c_{\min} > 0$ such that $c_l \geq c_{\min}$ for all l , and $c_{\max} = 1$ trivially.

From population to empirical. For N data points sampled iid from the uniform measure on S^{d-1} , the empirical kernel matrix $K_{ij}^{(N)} = K(x_i, x_j)$ satisfies, with high probability, $\|K^{(N)} - K\|_{op} = O(N^{-1/2})$ by standard matrix-Bernstein concentration [Tropp, 2015]. The same applies to $\Sigma^{(1),(N)}$. Their eigenvectors converge in the appropriate sense (Davis–Kahan) to projections of spherical harmonics, so the alignment claim of the proposition holds up to $O(N^{-1/2})$ deviations at finite N .

Combining for the diagonal-entry claim. Writing the population kernels in their shared eigenbasis U_∞ (spherical harmonics) and using $\Sigma^{(1)} = K \cdot \text{diag}(c_l)$ in that basis, the empirical $\tilde{G}_0[i, i]$ at finite N satisfies $\tilde{G}_0[i, i] = c_{l(i)} \lambda_i + O(N^{-1/2}, M^{-1/2})$ where $l(i)$ is the spherical-harmonic frequency level of the i -th empirical eigenvector. With $c_l \in [c_{\min}, c_{\max}]$ this proves the proposition. $\square$

B Implementation details

All experiments use NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ per layer. We integrate the kBM ODE with classical RK4 at the same step size as the network’s training step,

so gradient-flow time is matched. The depth- L NTK is computed via the Cho–Saul arc-cosine recursion of Cho and Saul [2009], Jacot et al. [2018].

For the Fashion-MNIST experiment we use four classes (T-shirt, Trouser, Pullover, Dress) and form 1000 triplets per seed by sampling (anchor, positive, negative) with same/different class labels. Inputs are PCA-reduced to 64 dimensions and unit-normalized before being passed through the network.

For the end-to-end CIFAR-10 experiment we train a 3-block CNN (32-32-64 channels, 32-dim projection head, $\sim 104\text{k}$ params) on $N = 100$ images with triplet loss. The empirical NTK at initialisation is computed by per-output-feature backward passes (`runners/cifar10_end2end.py`). The preconditioned variant uses $(K^{\text{emp}} + \varepsilon I)^{-1}$ with $\varepsilon = 10^{-2}$.

Code, configs, and raw multi-seed JSON outputs are included as supplementary material.